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Batch Alcoholic Fermentation Modelling by Simultaneous Integration of Growth and Fermentation Equations

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ABSTRACT

The modelling of batch growth and fermentation data presents important disadvantages when numerically calculated specific rates are used. In this paper, the results obtained using a different technique, based on the integration of the differential rate equations in the modelling of batch alcoholic fermentation with Saccharomyces cerevisiae *at different temperatures and initial sugar concentrations, are presented, and the use of both techniques is discussed.*

Key words: alcoholic fermentation, yeast growth, kinetic modelling, *Saccharomyces cerevisiae.*

NOMENCLATURE

K_d	self-inhibition constant for growth (h^{-1})
K_s	constant in the substrate term for growth (g dm^{-3})
K_s'	constant in the substrate term for ethanol production (g dm^{-3})
K_p	constant of growth inhibition by ethanol (g dm^{-3})
K_p'	constant of fermentation inhibition by ethanol (g dm^{-3})
R_{sx}	ratio of cell produced per glucose consumed for growth
R_{sp}	ratio of ethanol produced per glucose consumed for fermentation
S	glucose concentration (g dm^{-3})
P	ethanol concentration (g dm^{-3})
X	cell concentration (g (d.w.) dm^{-3})
Y_{sp}	fermentation yield (g ethanol g^{-1} glucose)

Greek letters

μ specific growth rate (h^{-1})
μ_m maximum specific growth rate (h^{-1})
ν specific fermentation rate (h^{-1})
ν_m maximum specific fermentation rate (h^{-1})

1 INTRODUCTION

Alcoholic fermentation by yeast is a widely studied reaction, either from the basic or applied research point of view, especially since ethanol is considered an alternative energy source and is used as a partial substitute for gasoline as a fuel. However, the basic understanding of alcoholic fermentation has not completely been established. The quantification of intracellular ethanol accumulation,[1,2] the inhibitory effects of intracellular and extracellular ethanol on growth and fermentation activity,[3,4] the possible inhibitory effects of secondary products other than ethanol,[5–7] the effect of CO_2[8] and the quantification of the oxygen level necessary in the broth,[9,10] are several areas where more research is needed. However, the interest in the industrial application of alcoholic fermentation made evident the need to build growth and fermentation models for application in fermenter design. For this reason many kinetic models have been proposed in the literature, generally being modifications of the Monod equation taking into account the ethanol inhibitory effect. These are reviewed in Refs 4 and 11, for example. Most models have been applied to different experimental data, obtained either in batch or continuous culture.[4,10–22] To determine the values of the model parameters, their equations are usually written in the form:

$$\mu=f(S,\ P) \text{ with } \mu=\frac{1}{X}\frac{dX}{dt} \tag{1}$$

$$\nu=f(S,\ P) \text{ with } \nu=\frac{1}{X}\frac{dP}{dt} \tag{2}$$

In the case of batch fermentations, the calculation of μ and ν from the experimental growth and fermentation curves requires: first, their numerical fitting (using, for example, a polynomial approach) and second, the numerical calculation of the derivatives from the fitted curve. In addition to the accuracy problems associated with the use of numerical derivation, the fact that the rate values, especially for growth, have large variations in short time intervals (e.g. μ changes from its maximal value to nearly zero) causes the derivative values to be very sensitive to the numerical fitting of the experimental data. In consequence, little variation in that fitting can give significant changes in the calculated rate values, μ and ν, and therefore in the evaluation of the model parameters, which is done by the nonlinear least squares method.

To avoid these problems, the modelling can be done by direct simultaneous integration of the differential equations for growth and fermentation

$$\frac{dX}{dt}=f_1(X, S, P) \quad (3)$$

$$\frac{dP}{dt}=f_2(X, S, P) \quad (4)$$

$$\frac{dS}{dt}=f_3\left(\frac{dX}{dt}, \frac{dP}{dt}\right) \quad (5)$$

the results obtained in the integration are compared with the experimental data and through an optimization routine the values of the model parameters are changed until the error between experimental and calculated data is minimal. In the present study, the modelling of several batch alcoholic fermentation experiments has been evaluated by two different models using the integration strategy, which in the following will be referred to as method 1. The results are compared with those obtained using the conventional technique described above (using numerically calculated derivatives) which in the following will be referred to as method 2.

2 MATERIALS AND METHODS

2.1 Microorganisms

The yeast strain used in this work was isolated from a high grade wine and was identified as *Saccharomyces cerevisiae* Hansen.

2.2 Analytical methods

Glucose and ethanol were determined by HPLC as previously reported.[23]

2.3 Cell concentration measurement

Cell concentration was determined by measuring the absorbance at 650 nm in a Perkin–Elmer (Uberlinger, FRG) Lambda-5 spectrophotometer and calculating the concentration in g dry weight dm^{-3} by means of a calibration plot previously established.

2.4 Batch fermentation experiments

These were carried out in 1 dm^{-3} ehrlenmeyer flasks in a shaking bath at constant temperature. The fermentation medium composition was: glucose (75–200 g dm^{-3}); yeast extract (2 g dm^{-3}); NH_4Cl (1·3 g dm^{-3}); $MgSO_4$ (0·82 g dm^{-3}); KH_2PO_4 (2 g dm^{-3}); sodium citrate (1·1 g dm^{-3}) and citric acid (1·5 g dm^{-3}) (pH=4). The initial volume of medium was 600 cm^3 and a 2% by volume inoculum grown overnight was used. Experiments were performed at three temperatures: 25, 30 and 35°C; and four initial sugar levels in the medium: 75, 100, 150 and 200 g dm^{-3}. Cell, glucose and ethanol concentrations were followed until total glucose consumption.

3 MATHEMATICAL METHODS

3.1 Method 1: integration approach

In order to do the modelling through the methodology proposed in this work, the numerical simultaneous integration of eqns (3) to (5) was done using the Runge–Kutta method[24] and the results were compared directly to the experimental data using a Nelder–Mead optimization routine[25] coupled with the integration routine. The model parameters were changed in the optimization routine and fed again to the integration step until minimal error between experimental and integrated values was achieved.

3.2 Method 2: conventional approach

When the modelling was carried out in the form of the eqns (1) and (2), first the experimental growth and fermentation data were fitted with a curve using the polynomial Chebyshev series method (NAG library routines)[26] ensuring the continuity in their first derivative. Then the specific rates were calculated. The evaluation of the model parameter was done by minimizing the error between these values and those predicted by the model, using the nonlinear least squares method of Marquardt.[27]

In both cases, the residue between experimental and calculated data was evaluated by:

$$\text{Residue}=\sum_{i=1}^{n}(C_{\text{exp}}-C_{\text{cal}})^2 \tag{6}$$

4 RESULTS AND DISCUSSION

The results given below are divided basically into three parts. In the first and second, data obtained in the fermentation experiments at 30°C are used to compare methods 1 and 2; employing, in both cases, two models selected from those commonly used in the literature. In the third part, the modelling of data at 25 and 35°C is carried out with the best method and model found in the previous stages.

4.1 Method 1

Four experiments were performed at 30°C, using different levels of glucose in the medium: 75, 100, 150 and 195 g dm^{-3}. The experimental variation of yeast (X), glucose (S) and ethanol (P) concentration corresponded to method 1 described in Section 2 using the following two models.

4.1.1 Model A

Proposed by Aiba *et al.*[13]

$$\frac{dX}{dt}=\mu_m\cdot X\cdot\frac{S}{K_s+S}\cdot\exp\left(-K_pP\right) \tag{7}$$

$$\frac{dP}{dt}=\nu_m \cdot X \cdot \frac{S}{K_s'+S} \cdot \exp(-K_p'P) \tag{8}$$

$$\frac{dS}{dt}=-\frac{1}{R_{sx}} \cdot \frac{dX}{dt}-\frac{1}{R_{sp}} \cdot \frac{dP}{dt} \tag{9}$$

where: R_{sx}, R_{sp} are yield factors defined as the ratios of cell and ethanol produced per the corresponding amount of glucose used for growth or ethanol production respectively (in consequence, the expected values for R_{sp} will be greater than thôse of Y_{sp}, which are around 0·43 g ethanol g^{-1} glucose).

4.1.2 Model B

With a term that represents the non-competitive inhibition by ethanol, used by various authors[4,15] and another reflecting biomass self-inhibition during growth:

$$\frac{dX}{dt}=\mu_0 \cdot X \cdot \frac{S}{K_s+S} \cdot \frac{K_p}{K_p+P}-K_d X \tag{10}$$

$$\frac{dP}{dt}=\nu_m \cdot X \cdot \frac{S}{K_s'+S} \cdot \frac{K_p'}{K_p'+P} \tag{11}$$

$$\frac{dS}{dt}=-\frac{1}{R_{sx}} \cdot \frac{dX}{dt}-\frac{1}{R_{sp}} \cdot \frac{dP}{dt} \tag{12}$$

where the symbols have the same meaning as those in previous equations, except the fact that μ_0 is now greater than μ_m and K_d is the self-inhibition constant (h^{-1}).

The results obtained with both models for the fermentation experiments at 30°C are shown in Figs 1 to 4, where experimental data are represented by points and the model curves by lines. Also, the values of the parameters obtained in the modelling and the residues between experimental and calculated data are presented in Tables 1 and 2. The fitting of the predicted curves to experimental data is very satisfactory for both models, in spite of the complexity involved in solving simultaneously the three variables, X, S and P. Neverthelesss, the behaviour of model B seems slightly better than that of model A because it gives a better fit in the transition from the exponential growth phase to the stationary phase, which is a particularly interesting zone in the growth curve. Moreover, model A predicts a constant concentration of cells once the stationary phase is reached, while model B describes a decrease in cell number after the stationary phase, which conceptually is more correct because the number of dead cells becomes relatively important in that part.

In fact, the minimal residues beween calculated and experimental data are lower for model B, reflecting its better behaviour. Moreover, the general validity of a model is normally reflected in obtaining the same value for its parameters for different initial sugar concentrations in the medium. This criterion has been

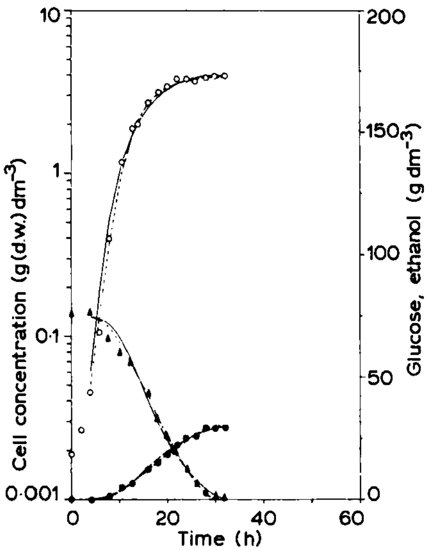

Fig. 1. Modelling of growth and fermentation data at 30°C and S_0=75 g dm^{-3}. ○, cell concentration (g dm^{-3}); ▲, glucose concentration (g dm^{-3}); ●, ethanol concentration (g dm^{-3}); ----, model A; ——, model B.

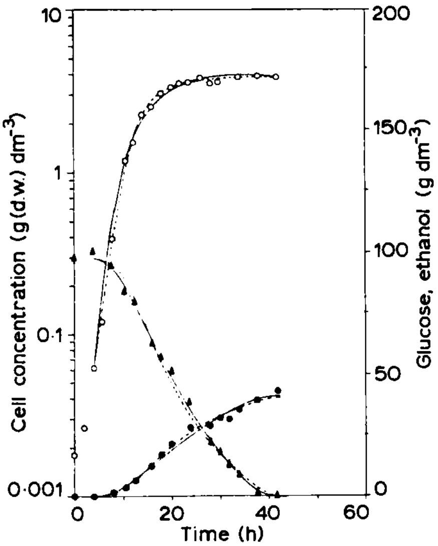

Fig. 2. Modelling of growth and fermentation data at 30°C and S_0=100 g dm^{-3}. ○, cell concentration (g dm^{-3}); ▲, glucose concentration (g dm^{-3}); ●, ethanol concentration (g dm^{-3}); ----, model A; ——, model B.

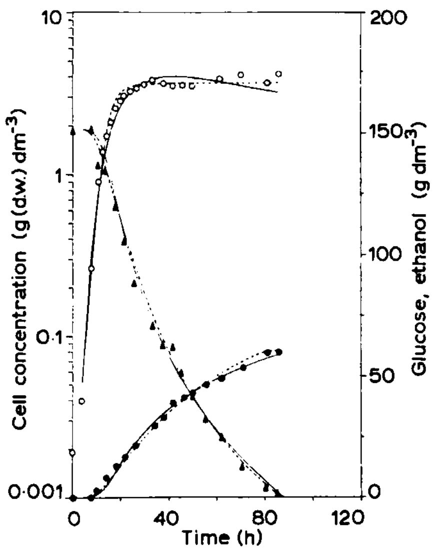

Fig. 3. Modelling of growth and fermentation data at 30°C and S_0=150 g dm^{-3}. ○, cell concentration (g dm^{-3}); ▲, glucose concentration (g dm^{-3}); ●, ethanol concentration (g dm^{-3}); ----, model A; ——, model B.

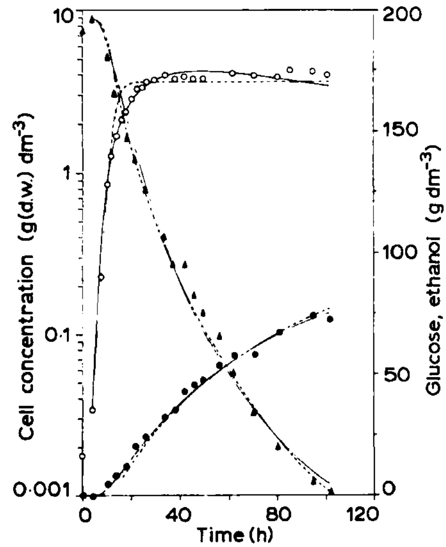

Fig. 4. Modelling of growth and fermentation data at 30°C and S_0=195 g dm^{-3}. ○, cell concentration (g dm^{-3}); ▲, glucose concentration (g dm^{-3}); ●, ethanol concentration (g dm^{-3}); ----, model A; ——, model B.

TABLE 1
Value of the Kinetic Parameters and Residues Obtained in the Modelling by Simultaneous Integration with the Model A for the Experiments at 30°C

S_0 (g dm^{-3})	μ_m (h^{-1})	K_s (g dm^{-3})	K_p (g dm^{-3})	ν_m (h^{-1})	K_s' (g dm^{-3})	K_p' (g dm^{-3})	R_{sx}	R_{sp}	Residue X	Residue S	Residue P
75	0·556	1·03	0·139	1·79	1·680	0·070	0·607	0·435	0·05	2·6	2·8
100	0·516	1·01	0·138	1·51	0·852	0·053	0·930	0·431	0·08	73·0	16·0
150	0·480	0·14	0·222	0·72	0·682	0·032	1·130	0·406	0·72	82·0	56·0
195	0·593	0·87	0·307	0·54	0·933	0·018	0·145	0·454	0·59	305·0	240·0

TABLE 2
Value of the Kinetic Parameters and Residues Obtained in the Modelling by Simultaneous Integration with the Model B for the Experiments at 30°C

S_0 (g dm^{-3})	μ_0 (h^{-1})	K_s (g dm^{-3})	K_p (g dm^{-3})	K_d (h^{-1})	ν_m (h^{-1})	K_s' (g dm^{-3})	K_p' (g dm^{-3})	R_{sx}	R_{sp}	Residue X	Residue S	Residue P
75	0·522	0·210	2·42	0·0078	1·36	1·68	7·29	0·299	0·448	0·01	2·7	1·4
100	0·723	0·236	2·24	0·0098	1·78	1·64	7·10	0·299	0·484	0·09	9·1	7·1
150	0·729	0·203	2·15	0·0094	0·86	2·29	7·25	0·325	0·477	0·15	77·0	16·0
195	0·880	0·539	1·21	0·0052	1·54	2·09	6·93	0·197	0·483	0·18	76·0	51·0

fulfilled for the two tested models, within a certain scattering, however, the results for model B are again slightly better than those for model A; especially the greater variation in parameters K_s, ν_m and K_s' in model A.

Consequently, model B is more appropriate for the description of batch growth and fermentation of the yeast strain tested. It is also important to note that the model describes accurately the evolution of cell, sugar and ethanol concentrations from the beginning of the exponential phase to the stationary phase. Also, the values of the parameters obtained are in agreement with those reported previously. For example, the values given by Hoppe and Hansford[10] modelling the growth of *Saccharomyces cerevisiae* ATCC 4126 from continuous runs are: μ_m=0·6 h^{-1}; K_s=3·3 g dm^{-3} and K_p=5·2 g dm^{-3}.

4.2 Method 2

When the data obtained at 30°C are treated with the method using directly the derivatives obtained numerically from the curves as indicated in Section 2, the model equations have the following form.

4.2.1 Model A'

$$\mu=\mu_m\cdot\frac{S}{K_s+S}\cdot\exp(-K_pP) \tag{13}$$

$$\nu=\nu_m\cdot\frac{S}{K_s'+S}\cdot\exp(-K_p'P) \tag{14}$$

4.2.2 Model B′

$$\mu=\mu_0\cdot\frac{S}{K_s+S}\cdot\frac{K_p}{K_p+P}-K_d \tag{15}$$

$$\nu=\nu_m\cdot\frac{S}{K_s'+S}\cdot\frac{K_p'}{K_p'+P} \tag{16}$$

The results obtained are presented in Tables 3 to 6. Although the values of the residues are lower (mainly because the absolute values of the data are lower than those in the previous case and less points are being modelled—only one curve instead of three) the fitting of the model to the calculated value is sensibly less accurate, especially for low times, where in fact the changes are more pronounced (see Figs 5 and 6, where the results for one case are given as an example).

TABLE 3
Results of the Modelling of the Specific Growth Rates with Model A′ for the Experiments at 30°C

S_0 (g dm^{-3})	μ_m (h^{-1})	K_s (g dm^{-3})	K_p (g dm^{-3})	*Residue*
75	0·457	7·98	0·118	1·1 10^{-4}
100	0·402	1·10	0·123	2·6 10^{-4}
150	0·450	0·10	0·150	5·9 10^{-4}
195	0·699	0·10	0·153	4·8 10^{-4}

TABLE 4
Results of the Modelling of the Specific Fermentation Rates with Model A′ for the Experiments at 30°C

S_0 (g dm^{-3})	ν_m (h^{-1})	K_s' (g dm^{-3})	K_p' (g dm^{-3})	*Residue*
75	1·34	0·61	0·048	3·0 10^{-3}
100	2·15	0·47	0·059	2·5 10^{-3}
150	1·05	0·35	0·055	1·3 10^{-3}
195	2·35	0·53	0·061	4·6 10^{-3}

TABLE 5
Results of the Modelling of the Specific Growth Rates with Model B′ for the Experiments at 30°C

S_0 (g dm^{-3})	μ_0 (h^{-1})	K_s (g dm^{-3})	K_p (g dm^{-3})	K_d (h^{-1})	*Residue*
75	0·560	0·1	3·80	0·0162	1·9 10^{-5}
100	0·800	0·1	2·00	0·0108	1·0 10^{-4}
150	1·140	0·1	1·15	0·0104	6·0 10^{-6}
195	2·580	0·1	0·73	0·0159	8·2 10^{-4}

TABLE 6
Results of the Modelling of the Specific Fermentation Rates with Model B′ for the Experiments at 30°C

S_0 ($g\ dm^{-3}$)	v_m (h^{-1})	K_s' ($g\ dm^{-3}$)	K_p' ($g\ dm^{-3}$)	Residue
75	3·15	0·75	2·67	$2{\cdot}6\ 10^{-3}$
100	2·51	1·95	3·64	$2{\cdot}2\ 10^{-3}$
150	1·84	5·61	7·97	$2{\cdot}3\ 10^{-2}$
195	4·04	0·15	1·88	$6{\cdot}8\ 10^{-3}$

TABLE 7
Value of the Kinetic Parameters and Residues Obtained in the Modelling by Simultaneous Integration with the Model B for the Experiments at 25°C

S_0 ($g\ dm^{-3}$)	μ_0 (h^{-1})	K_s ($g\ dm^{-3}$)	K_p ($g\ dm^{-3}$)	K_d (h^{-1})	v_m (h^{-1})	K_s' ($g\ dm^{-3}$)	K_p' ($g\ dm^{-3}$)	R_{sx}	R_{sp}	Residue		
										X	S	P
75	0·573	0·121	2·12	0·0075	1·36	1·72	7·715	0·266	0·539	0·02	15·8	3·4
95	0·549	0·148	2·11	0·0050	1·28	1·71	7·160	0·258	0·525	0·20	30·8	7·7
144	0·553	0·199	1·62	0·0040	1·14	2·73	6·75	0·235	0·512	0·53	65·0	25·0
180	0·639	0·405	1·16	0·0027	1·19	2·38	6·93	0·594	0·465	0·96	54·6	55·7

Moreover, the variation in the obtained value of the parameters for different S_0 values is now more important especially as the parameters K_s and K_s' are very insensitive to the fitting, being in most cases the minimum allowed value in the optimization routine, 0·1 g dm^{-3}. These facts, together with the general drawbacks pointed out in the introduction, indicate that the simultaneous integration procedure is more appropriate to carry out the modelling.

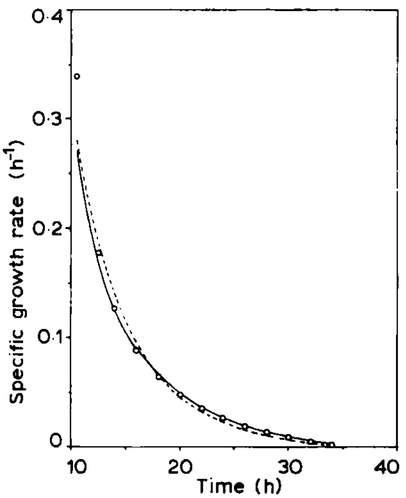

Fig. 5. Modelling of specific growth rates at 30°C and S_0=150 g dm^{-3}. -----, model A′; ——, model B′.

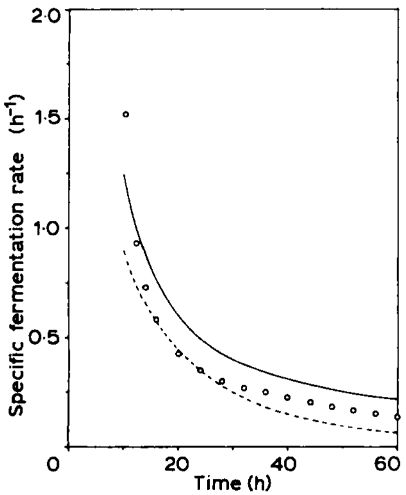

Fig. 6. Modelling of specific fermentation rates at 30°C and S_0=150 g dm^{-3}. -----, model A′; ——, model B′.

TABLE 8
Value of the Kinetic Parameters and Residues Obtained in the Modelling by Simultaneous Integration with the Model B for the Experiments at 35°C

S_0 $(g\ dm^{-3})$	μ_0 (h^{-1})	K_s $(g\ dm^{-3})$	K_p $(g\ dm^{-3})$	K_d (h^{-1})	ν_m (h^{-1})	K_s' $(g\ dm^{-3})$	K_p' $(g\ dm^{-3})$	R_{sx}	R_{sp}	Residue		
										X	S	P
77	0·458	0·399	2·412	0·0056	1·325	0·220	7·84	0·447	0·466	0·3	68·0	30·0
104	0·445	0·219	2·262	0·0071	1·275	0·270	7·22	0·954	0·467	1·0	77·8	20·6
157	0·448	0·636	2·380	0·0064	1·511	0·127	7·99	0·863	0·454	0·7	215·0	180·0

Finally, using method 2, the application of model B has been extended to the data obtained in the fermentation experiments at 25 and 35°C, also giving an accurate fitting of the model to experimental data, very similar to that for 30°C. The results are presented in Tables 7 and 8, where it can be seen that the value of the parameters obtained is also quite similar for different runs at the same temperature. Thus, the behaviour of both the methodology and the model is also satisfactory for the experimental data at 25 and 35°C.

5 CONCLUSIONS

The various numerical disadvantages which appear when data of yeast growth and fermentation batch experiments are modelled using calculated specific rates (method 2 in this paper) can be avoided by integrating directly the differential equations for cell, substrate and product concentrations (method 1 in this paper). This fact has been tested with two different models among those used in the literature and one of them (eqns (10) to (12)) has been selected as the most appropriate to describe the results obtained at different temperatures and initial sugar concentrations with the yeast strain used in the present study.

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